

1-8B Llama vs. Mamba2 vs. GLA2 vs. GDN5 || Tasks: Im_eval (left), JRT (middle), multi-hop (right)

8B 1.0T(4096) Llama - original - Ir0.003	-66.5%	66.4%	81.2%	78.6%	70.8%	83.3%	56.0%	48.3%	80.8%	89.0%	46.2%	37.4%	65.5%	60.6%	9.62	4.60	63.7%	80.5%	87.6%	71.6%	73.9%	42.3%	45.4%	54.7%	69.2%	74.4%	73.6%	34.8%	56.4%	100.0%	99.5%	99.1%	93.6%	90.1%	80.4%	2.6%	0.6%	54.1%	37.2%	30.4%	29.9%	29.9%	29.8%	1.7%	0.7%
8B 1.0T(4096) Llama - original - Ir0.002	-66.5%	66.3%	80.3%	78.6%	71.1%	83.3%	55.8%	47.4%	80.9%	92.0%	45.0%	39.0%	64.3%	60.2%	9.60	4.67	64.3%	81.4%	89.0%	73.4%	81.1%	42.8%	45.7%	54.1%	68.3%	72.2%	72.7%	34.6%	56.0%	100.0%	99.9%	99.4%	97.4%	95.0%	88.8%	0.6%	0.5%	42.9%	35.9%	33.1%	30.9%	32.3%	32.4%	0.5%	0.0%
8B 1.0T(4096) Llama - Canon-ABCD - Ir0.003	-67.5%	68.2%	80.6%	79.5%	72.9%	83.4%	57.8%	46.7%	83.5%	89.0%	47.4%	39.5%	66.7%	62.1%	9.21	4.27	64.2%	82.2%	85.1%	74.9%	73.1%	42.8%	47.1%	53.6%	70.7%	74.4%	72.4%	34.3%	59.5%	100.0%	99.1%	99.4%	98.7%	96.7%	92.8%	26.7%	9.7%	50.8%	39.5%	36.3%	33.8%	32.1%	32.5%	10.0%	3.3%
8B 1.0T(4096) Llama - Canon-ABCD - Ir0.002	-67.1%	69.4%	80.6%	79.4%	71.9%	84.1%	57.3%	46.6%	82.0%	87.0%	45.2%	38.8%	67.4%	62.4%	9.19	4.20	64.3%	77.7%	85.4%	75.9%	74.1%	43.0%	47.2%	54.6%	71.0%	74.9%	73.6%	36.0%	58.7%	100.0%	99.9%	99.7%	98.2%	98.2%	94.6%	12.2%	5.3%	39.0%	42.5%	37.0%	34.5%	35.1%	33.9%	5.5%	4.5%
8B 1.0T(4096) Mamba2 - conv - Ir0.003	-67.1%	68.1%	80.6%	79.2%	71.8%	83.8%	57.8%	48.2%	81.8%	90.0%	47.4%	37.7%	64.4%	61.3%	9.36	4.32	60.2%	66.8%	79.3%	62.7%	76.6%	40.5%	44.5%	51.8%	69.6%	73.3%	72.9%	31.3%	53.2%	100.0%	95.2%	84.7%	59.2%	40.2%	21.4%	10.3%	5.6%	33.3%	33.2%	32.8%	28.2%	26.2%	20.6%	13.7%	9.1%
8B 1.0T(4096) Mamba2 - conv - Ir0.002	-66.9%	68.3%	81.5%	79.5%	72.8%	84.3%	56.5%	48.3%	81.5%	90.0%	45.6%	37.6%	61.8%	61.6%	9.38	4.30	61.1%	70.8%	80.1%	66.8%	78.2%	38.7%	44.2%	52.5%	67.9%	73.6%	72.9%	33.3%	54.0%	99.9%	95.0%	88.5%	67.9%	49.9%	27.5%	12.2%	4.4%	33.5%	34.7%	31.3%	28.9%	25.8%	21.3%	13.8%	7.2%
8B 1.0T(4096) Mamba2 - Canon-AbCD - Ir0.003	-66.8%	67.3%	80.7%	79.4%	70.8%	84.4%	58.2%	47.4%	81.1%	91.0%	45.0%	37.5%	64.9%	61.4%	9.33	4.60	59.4%	66.9%	79.7%	63.6%	71.2%	40.0%	42.9%	50.8%	67.4%	73.3%	72.8%	32.3%	52.0%	100.0%	92.1%	84.4%	62.1%	39.1%	21.4%	11.0%	2.0%	38.2%	32.5%	33.0%	29.8%	25.8%	21.2%	10.1%	2.0%
8B 1.0T(4096) Mamba2 - Canon-AbCD - Ir0.002	-66.7%	67.6%	80.5%	79.7%	71.3%	83.9%	57.5%	47.4%	80.2%	90.0%	48.4%	37.3%	62.2%	61.6%	9.32	4.41	59.6%	65.2%	74.0%	67.1%	69.5%	40.5%	44.6%	52.7%	69.9%	74.2%	73.6%	31.7%	52.2%	99.9%	94.9%	84.6%	56.1%	35.9%	18.9%	7.3%	2.7%	37.9%	33.6%	30.2%	29.2%	23.5%	18.7%	8.7%	2.1%
8B 1.0T(4096) GLA5 - conv - Ir0.003	-67.4%	67.4%	81.1%	79.5%	72.3%	84.6%	57.3%	47.5%	80.6%	91.0%	47.4%	39.7%	67.0%	61.3%	9.47	4.63	60.0%	66.2%	77.5%	66.2%	75.7%	40.1%	43.4%	51.0%	68.0%	73.3%	72.6%	31.9%	54.4%	100.0%	97.3%	88.6%	64.8%	48.1%	32.9%	24.7%	18.5%	33.6%	31.6%	28.7%	26.7%	25.4%	22.3%	19.5%	17.1%
8B 1.0T(4096) GLA5 - conv - Ir0.002	-66.8%	67.4%	81.1%	79.0%	72.2%	83.8%	57.8%	46.6%	80.0%	92.0%	46.6%	39.3%	61.8%	61.3%	9.48	4.77	60.8%	65.9%	81.7%	66.3%	78.0%	40.8%	43.0%	51.5%	69.9%	73.9%	72.6%	33.3%	52.3%	100.0%	97.3%	90.2%	65.1%	49.9%	34.6%	26.3%	18.8%	35.6%	28.3%	26.8%	27.5%	22.8%	24.1%	17.7%	16.3%
8B 1.0T(4096) GLA5 - Canon-AbCD - Ir0.003	-67.5%	69.7%	80.8%	79.8%	73.3%	83.6%	58.4%	47.1%	80.7%	88.0%	46.6%	38.1%	68.6%	62.6%	9.23	4.23	59.6%	63.0%	74.3%	64.7%	73.0%	40.8%	43.7%	53.9%	68.4%	72.6%	71.6%	32.8%	56.4%	100.0%	97.9%	86.8%	55.0%	36.7%	26.7%	21.3%	15.6%	33.8%	30.1%	31.9%	27.7%	31.2%	26.7%	25.4%	22.2%
8B 1.0T(4096) GLA5 - Canon-AbCD - Ir0.002	-67.7%	68.2%	81.0%	80.0%	72.3%	83.8%	59.0%	47.3%	81.2%	92.0%	47.2%	37.9%	67.0%	63.0%	9.22	4.29	61.6%	68.2%	81.1%	65.2%	75.8%	40.6%	44.6%	53.6%	73.7%	73.6%	73.6%	31.9%	56.3%	100.0%	98.2%	90.7%	66.6%	49.4%	36.5%	25.4%	20.6%	36.6%	31.2%	32.4%	31.0%	31.3%	28.9%	26.0%	21.1%
8B 1.0T(4096) GDN2 - conv - Ir0.003	-67.6%	68.5%	81.3%	79.0%	72.8%	83.8%	58.4%	47.0%	80.8%	92.0%	46.8%	38.6%	67.8%	63.1%	9.38	4.23	60.9%	67.0%	79.4%	66.8%	75.6%	39.8%	44.6%	53.0%	71.6%	73.3%	73.4%	32.2%	53.7%	100.0%	96.7%	89.1%	61.1%	40.0%	28.3%	19.4%	14.0%	33.9%	33.1%	31.7%	32.7%	30.3%	26.9%	23.9%	20.7%
8B 1.0T(4096) GDN2 - conv - Ir0.002	-67.6%	68.1%	81.0%	79.0%	74.7%	85.3%	59.5%	46.7%	79.2%	90.0%	47.6%	38.6%	66.0%	62.8%	9.40	4.33	60.6%	65.8%	79.3%	66.2%	74.8%	41.0%	46.0%	52.0%	68.7%	72.3%	73.0%	34.5%	54.2%	100.0%	95.4%	87.8%	62.2%	46.1%	33.3%	25.3%	18.6%	33.7%	34.0%	32.0%	30.7%	29.7%	26.2%	23.3%	20.4%
8B 1.0T(4096) GDN2 - Canon-AbCD - Ir0.003	-67.7%	68.5%	81.2%	79.5%	72.0%	85.3%	58.0%	47.9%	80.0%	90.0%	48.0%	38.6%	67.5%	63.0%	9.20	4.46	61.5%	72.2%	82.1%	66.1%	73.4%	41.2%	45.2%	51.9%	71.6%	72.5%	73.3%	32.9%	55.5%	100.0%	97.6%	89.3%	60.9%	40.6%	31.9%	25.6%	20.9%	34.4%	31.2%	30.6%	32.2%	31.4%	27.8%	27.2%	22.2%
8B 1.0T(4096) GDN2 - Canon-AbCD - Ir0.002	-67.9%	68.8%	81.9%	79.6%	73.2%	84.5%	59.8%	48.2%	81.7%	90.0%	45.4%	38.9%	66.4%	64.0%	9.23	4.25	61.5%	72.3%	81.1%	68.8%	75.0%	40.7%	44.7%	53.1%	69.4%	73.2%	73.9%	31.7%	54.5%	100.0%	97.2%	89.5%	68.2%	56.1%	50.2%	39.5%	38.7%	33.9%	33.7%	31.0%	29.2%	28.7%	26.8%	25.0%	25.3%
3B 1.0T(4096) Llama - original - Ir0.003	-61.9%	63.8%	78.2%	74.3%	65.0%	79.4%	49.8%	44.4%	73.1%	88.0%	44.8%	37.8%	52.7%	53.2%	11.45	5.47	58.0%	77.4%	88.1%	59.3%	63.8%	38.8%	41.7%	52.0%	62.0%	72.2%	68.6%	30.3%	41.8%	99.9%	98.1%	93.9%	91.4%	80.8%	53.7%	0.0%	0.2%	34.2%	32.2%	29.4%	32.1%	24.1%	27.5%	0.0%	0.0%
3B 1.0T(4096) Llama - original - Ir0.002	-61.7%	63.1%	79.6%	74.3%	65.9%	79.3%	50.3%	44.9%	73.5%	86.0%	43.4%	38.9%	50.3%	52.3%	11.45	5.66	56.8%	77.6%	87.6%	57.0%	56.6%	38.4%	38.8%	51.8%	62.3%	71.2%	67.7%	29.3%	42.6%	100.0%	99.6%	98.2%	94.7%	89.3%	54.6%	0.0%	0.1%	35.2%	36.4%	34.5%	33.9%	30.3%	30.5%	0.1%	0.1%
3B 1.0T(4096) Llama - Canon-ABCD - Ir0.003	-63.2%	64.9%	79.1%	74.9%	66.9%	79.3%	51.6%	45.4%	75.1%	83.0%	44.8%	38.4%	62.7%	55.2%	11.09	5.09	58.5%	78.5%	85.3%	61.2%	63.4%	41.0%	44.4%	52.3%	63.7%	71.7%	67.2%	29.2%	44.3%	100.0%	99.4%	99.3%	97.0%	96.1%	93.6%	70.2%	0.2%	33.6%	33.8%	31.2%	32.7%	28.4%	30.3%	24.6%	0.0%
3B 1.0T(4096) Llama - Canon-ABCD - Ir0.002	-63.3%	64.5%	78.7%	75.0%	67.8%	79.8%	51.3%	45.5%	76.7%	87.0%	45.0%	37.6%	57.7%	56.1%	11.12	5.14	59.6%	81.8%	87.9%	63.5%	65.7%	40.3%	42.9%	53.3%	65.1%	71.9%	67.3%	30.9%	45.1%	100.0%	100.0%	99.6%	97.8%	96.8%	91.8%	46.2%	10.0%	37.2%	36.8%	32.8%	34.6%	31.3%	32.8%	24.5%	6.6%
3B 1.0T(12288) Mamba2 - conv - Ir0.005	-62.0%	63.1%	78.8%	74.9%	66.1%	80.5%	50.4%	45.7%	72.3%	86.0%	45.0%	37.1%	53.5%	52.7%	11.37	5.23	54.2%	63.6%	79.3%	51.8%	60.8%	36.5%	39.6%	47.1%	63.5%	70.9%	65.7%	26.8%	44.4%	99.8%	86.8%	73.8%	46.0%	29.9%	19.2%	11.0%	7.1%	34.4%	32.7%	31.0%	29.9%	26.1%	22.3%	19.7%	16.2%
3B 1.0T(12288) Mamba2 - conv - Ir0.003	-61.9%	62.6%	78.8%	75.2%	65.3%	79.8%	52.0%	44.9%	74.0%	85.0%	44.8%	36.8%	51.4%	53.5%	11.36	5.34	53.2%	67.8%	73.7%	47.9%	58.1%	35.3%	39.8%	47.6%	62.8%	69.4%	66.9%	26.6%	42.4%	100.0%	92.0%	84.8%	61.1%	41.7%	29.1%	20.1%	17.1%	33.5%	32.7%	30.0%	28.9%	27.1%	24.5%	21.3%	17.3%
3B 1.0T(12288) Mamba2 - conv - Ir0.002	-60.8%	62.4%	78.3%	73.9%	66.1%	79.0%	49.9%	44.6%	72.0%	85.0%	44.4%	36.7%	47.4%	50.4%	11.56	5.52	53.6%	65.4%	76.3%	51.9%	58.8%	35.6%	39.8%	46.6%	62.2%	69.7%	65.8%	26.5%	44.2%	99.7%	83.7%	73.2%	47.8%	29.6%	21.0%	14.5%	12.7%	33.6%	34.1%	30.9%	28.3%	26.7%	23.3%	18.5%	14.8%
3B 1.0T(12288) Mamba2 - Canon-AbCD - Ir0.005	-63.2%	63.2%	79.5%	75.2%	68.0%	81.3%	53.2%	45.2%	74.3%	86.0%	45.4%	35.5%	58.7%	55.7%	11.22	5.04	54.7%	67.7%	75.4%	54.1%	63.2%	36.3%	41.7%	47.7%	61.3%	69.6%	67.7%	28.1%	43.5%	100.0%	88.8%	78.1%	49.9%	35.2%	24.3%	17.0%	14.6%	32.2%	35.1%	30.3%	29.7%	27.1%	22.7%	20.4%	16.7%
3B 1.0T(12288) Mamba2 - Canon-AbCD - Ir0.003	-62.9%	63.7%	80.2%	75.2%	67.3%	81.1%	50.3%	44.6%	74.7%	87.0%	44.0%	38.5%	55.2%	55.7%	11.22	5.14	55.6%	70.3%	79.1%	55.9%	65.1%	36.4%	41.8%	47.4%	62.8%	70.0%	66.4%	27.6%	44.6%	99.9%	93.8%	83.5%	58.4%	40.2%	30.7%	17.8%	13.8%	33.9%	31.2%	30.4%	25.7%	28.9%	26.1%	24.3%	19.0%
3B 1.0T(12288) Mamba2 - Canon-AbCD - Ir0.002	-61.7%	63.1%	79.1%	74.7%	66.1%	79.3%	51.1%	45.8%	71.5%	87.0%	44.6%	37.9%	48.6%	53.2%	11.43	5.41	53.3%	67.1%	77.3%	49.6%	56.8%	34.6%	39.1%	45.5%	62.5%	69.9%	68.6%	25.6%	43.3%	97.3%	91.5%	84.4%	60.8%	44.9%	30.5%	19.9%	14.5%	32.3%	31.4%	30.5%	32.9%	29.7%	26.8%	21.6%	19.6%
3B 1.0T(12288) GLA5 - conv - Ir0.005	-62.3%	62.2%	78.7%	74.9%	65.0%	80.1%	52.0%	45.7%	73.8%	89.0%	44.8%	36.8%	53.3%	53.7%	11.53	5.64	53.9%	68.4%	79.8%	46.7%	54.3%	35.8%	40.2%	45.4%	64.1%	70.1%	68.7%	27.6%	45.1%	100.0%	96.4%	89.5%	64.7%	43.2%	31.1%	19.9%	16.3%	34.4%	31.8%	32.4%	32.0%	26.4%	26.4%	23.2%	19.7%
3B 1.0T(12288) GLA5 - conv - Ir0.003	-62.5%	63.2%	79.3%	74.7%	67.7%	81.4%	52.8%	44.8%	73.3%	86.0%	44.2%	38.5%	52.9%	53.1%	11.47	5.50	54.8%	67.2%	81.1%	54.0%	59.6%	37.4%	40.9%	46.9%	63.4%	69.6%	67.6%	26.2%	44.1%	99.8%	96.6%	94.7%	73.3%	51.6%	36										